MK8

POWER SUPPLY

INSTRUCTION MANUAL

WARNING

When the Power Supply is, or has been energized, dangerous voltages exist which can cause serious injury or death.

WARNING

Read all warnings and procedures prior to start-up, troubleshooting or other work inside the power supply.

PP1223-1

Pillar Induction
21905 Gateway Road
Brookfield, Wisconsin 53045
Phone: (262) 317-5300

WARNINGS

WATER WARNING

DO NOT FILL WATER SYSTEM WITH DE-IONIZED WATER.

DO NOT USE ANY WATER ADDITIVES WITHOUT FIRST CONSULTING THE PILLAR INDUCTION SERVICE DEPARTMENT.

Pillar Induction
21905 Gateway Road
Brookfield, WI 53045
Weekdays: 262-317-5300
Nights: 414-225-6040

AC LINE PHASING WARNING

THIS UNIT REQUIRES PROPER AC LINE PHASING. PLEASE REFER TO PROPER CHAPTER WITHIN THIS MANUAL FOR ADDITIONAL INSTRUCTIONS.

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CHAPTER 1 INSTALLATION INSTRUCTIONS

Section I. Installation Instructions

A. Site Selection

The selection of a site for the power supply should be chosen with the following considerations:

1. Distance from KVA Source
   An economic advantage, as well as power advantage, will result in keeping the incoming power line length to a minimum. Less line drop will result if the 50/60 hertz lines are within a 200 foot run per phase. On the other hand, a very large KVA source (i.e. 10 x requirement) at short distances to the supply will impose little source impedance to the supply and may increase surge currents.
2. Distance from Load
   The distance between load and power supply should be kept as short as possible. An economic advantage will again result in minimizing the distance of transmission to the load. Long runs will increase inductance and also prevent power loss.
3. General Location
   The unit should be placed in a building to be protected from inclement weather. The ambient temperature must be kept above the freezing temperatures of the cooling lines. The power supply should be kept from high temperature sources. The equipment is capable of correct performance from an ambient operating temperature from above freezing to a maximum of 52° C. (125° F.)

B. Preparation for Installation

No base mounting is required for the power supply. Where multiple cabinets are used, they will be mounted on a common rail. A level area that is free from severe vibration should be selected. The operator's control panel may be ordered for remote installation or mounted in the power supply.

C. Input Power Connections

The MK8 Power Supply requires that the input three phase be properly phased when connected to the breaker. All references to "breaker" or "circuit breaker" in this manual refer to the non-automatic type circuit breakers (also referred to as a molded case switch). Before installation, local electrical codes should be reviewed for over-current or disconnect requirements. To assist in determining proper phasing, a phase sensing circuit is incorporated on the Rider Board located on the left-hand wall of the control cabinet. On the older Rider Board are two small lamps. One labeled IN PHASE, and the other labeled OUT PHASE. The input has been properly phased when only the IN PHASE lamp is lit.
There is an OUT PHASE LED on the feedback board also. The input is phased properly when this LED is not lit.

D. Cooling Water Requirements

An adequate water system should be made available to the power supply. The following is provided to assist in the selection of cooling equipment:

1. Connections and Flow
   Refer to installation drawings for flow and connections. Non-ferrous piping is suggested on all closed systems.
2. Protective Considerations
   The following protective equipment is provided for protection and comments added for clarity:
   a. Pressure switch to shut down the power supply if the differential pressure between inlet and outlet falls below 30 PSI. Water hammer should be avoided if operating near this level to avoid nuisance stopping.
   b. Water inlet temperature to shut down power supply if water rises to 46° C (115° F). For closed loop systems, a temperature switch is provided in each line for monitoring outlet rise. They, in turn, will energize and stop the supply at a nominal 65° C (150° F) +/- 5%. The lines are designed to operate considerably cooler than this under specified pressure. Upon energization of these circuits, check for a kinked hose, deposits, etc. restricting flow. A lamp for indicating purposes at each switch is provided and manual reset required, allowing for isolating any hot line due to deposits, build-up, etc. See Trouble-shooting for operation of lamps on 24V battery supply.
   c. Non-sacrificial targets to guard against electrolysis are provided. These should be periodically checked. The opposite end of the hose from the target connection should also be checked for deposits.
   d. Following are other considerations, which should be adhered to. Where reservoirs or open lines are used, care should be used in avoiding contaminates to the water. Items dropped into the water may lead to damage or blockage by deposits or harm to the system. The quality of water being used should dictate the frequency of inspection. Filters are suggested.
   e. When hose replacement is required, it is important to be sure that the replacement hose is <u>electrically non-conductive</u> and at least as long as the original hoses being replaced.
3. Dew Point
   The cooling water through the system should be kept above dew point to avoid condensation. When the unit is left off for long periods, no heat is generated within, and a water temperature below the prevailing dew point will act to condense moisture onto the equipment.
4. Drainage
   For maintenance purposes, one must place valves external to the power supply for shutting off the water source to the inlet and drain outlet (drain valve not required on site drain). A small purge valve is provided within the equipment for drainage of water from the lines. The user is to arrange for a convenient exit from the valve to outside the cabinet. Once initially drained by the purge valve to relieve pressure, one may remove individual lines and blow air back through the line exiting at the purge valve. This should be done during maintenance procedures or before storage to relieve the system of any trapped water.
   CAUTION: NO CHECK VALVE IS PLACED IN THE PURGE VALVE SO THAT WATER OR AIR MAY FLOW EITHER WAY.

CAUTION: IF STORING AT BELOW FREEZING TEMPERATURES, UNIT MUST BE FLUSHED WITH ANTI-FREEZE.

5. Quality of Water

Some questions generally arise to the quality of water for static systems. The following excerpt is taken for USA Standard C-34.2-1958 and is recommended. USE STANDARD PRACTICES AND REQUIREMENTS FOR SEMI-CONDUCTOR POWER RECTIFIERS C-34.2.

Section 7.5.2 – Heat Exchangers

Water-to-water heat exchangers are very satisfactory where adequate cooling water is available. They are generally operated at ground potential, but may be operated at the potential of either terminal. When differences of potential exist across insulating tubing or hose in the recirculating system, protection against electrolysis should be provided by the use of de-ionized water or electrolytic targets.

Water-to-air heat exchangers give very satisfactory service in areas where adequate supplies of suitable cooling water are not available.

Section 7.5.3 – Protection Against Freezing

In some installations, recirculating water-cooling systems require protection against freezing. Anti-freeze solutions or immersion heaters may be used. Where possibility of electrolysis exists and targets are not employed, commercial anti-freeze solutions containing corrosion inhibitors are generally not satisfactory since many of these inhibitors reduce the resistivity of the coolant.

Section 7.5.4 – Quality of Raw Water

Water for direct raw-water cooling system, where heat exchangers are not employed, should have the following purity:

1. A neutral or slightly alkaline reaction. (pH between 7.0 and 9.0)
2. A chloride content of not more than 20 parts per million. A nitrate content of not more than 10 parts per million. A sulfate content of not more than 100 parts per million.
3. A total solids content of not more than 250 parts per million.
4. A total hardness, as calcium carbonate, of not more than 250 parts per million.

Chemical analysis is not always available to assist in the appraisal of cooling water. In such cases, an electrical resistivity or conductivity measurement of the water will provide a satisfactory guide to the total amount of dissolved solids. Water having a resistivity of 2500 Ω centimeters minimum or a conductivity of 400micro mhos maximum, when measured at about 25°C, is usually satisfactory as a coolant. The approximate amount of dissolved solids can be determined by the equation:

$$\mathit{DISSOLVED\ SOLIDS}_{ppm} = \frac{640{,}000}{\mathit{Specific\ Resistivity}_{ohm-cm}}$$

Raw water insulating hose connections to the rectifier or ungrounded heat exchangers should be long enough to reduce the leakage current to a tolerable level, or electrolytic targets should be used at the hose fittings.

If antifreeze is required, pure ethylene glycol is the only recommended antifreeze with a mixture of no more than 40% ethylene glycol to 60% water in a pressurized cooling system. Consult Pillar Engineering for freeze protection on closed non-pressurized water systems.

Inlet cooling temperature regulation must be provided to insure inlet water temperature does not fall below dew point temperature to prevent condensation on critical electrical connections.

E. Remote Wiring and Interlocks

1. The power supply may be ordered with the operator's panel mounted in the cabinet or specified as remote. A minimum of control wiring is required with the panel cabinet mounted. Remote refers to having the meters and/or controls pre-mounted on a panel to be placed into the user's cabinet.
2. The customer is to use furnished drawings for wiring between the supply and load station. Standard practice of separate control conduit, isolation from power runs, good connections and twisting or shielding of wires where noted should be followed for wiring between power supply and remote operator panel, if any.
    a. A row of terminals is provided at the power supply.
    b. Customer interlocking provisions are made for the breaker holding circuit, low water pressure, high water temperature and START/STOP functions.
    c. Heat-on signal may be used by customer for heat "ON" feedback signal.
3. The following interlocking is incorporated in the system:
    a. The internal battery operated 24V DC supply in the master cabinet must be energized and all doors closed before engaging breaker.
    b. Failure to comply within specified limits of pressure (30 PSI differential), inlet waters temperature (46° C, 115° F), will prevent equipment from starting or will cause shutdown while in operation. Lamps are provided for an indication of fault at the control panel.

Section II. Recommended Procedures and Adjustments for Initial Starting

The power supply is sent from the factory tested and with adjustments factory set. The following should be adhered to in first starting the power supply. It assumes that the load has been attached to the power supply and has been set up for the approximate frequency.
CAUTION: HIGH VOLTAGES ARE PRESENT WITHIN THE CABINET. ONLY QUALIFIED PERSONNEL SHOULD BE ALLOWED TO WORK ON THE EQUIPMENT WITH DOORS OPEN. ALL SAFETY MEASURES SHOULD BE OBSERVED.

A. Inspection

Visually inspect unit to see that appropriate connections are made for the control, power in and power out. Check connections of buswork in power supply, terminal blocks, and control blocks before starting.

B. Application of Water

Connect cooling water to power supply and energize. Inspect cabinet to ensure that no leaks are present in individual lines at connection. Determine that adequate pressure exists. Pressure switches within the power supply are factory set to prevent starting the power supply if pressure is below 30 PSI differential between inlet and outlet. Closed loop systems have pressure gauges for determining the differential because of backpressure conditions.

C. Energizing Interlock Supply (24V DC)

Energize the 24V DC supply within the cabinet by pushing the enable switch.
CAUTION: DO NOT ATTEMPT TO TURN BREAKER ON WITHOUT ENERGIZING THE UV COIL.
The breaker ready lamp should be lit. If the breaker ready lamp is out, check all door interlocks. If all doors are closed, the breaker ready lamp will energize, indicating it is proper to close the breaker. This must be done while holding the enable switch. Once closed, power is supplied to the breaker DC under voltage coil and associated lights, wherein the enable switch may be released. If the breaker ready lamp fails to energize with doors closed, check the lamp and batteries. Opening any door of the cabinet will drop out the line breaker.

D. Energizing Power Circuit

Before closing the breaker, give the power supply a thorough visual and ohmmeter check (see Ohmmeter Procedure in the Maintenance Section). Ensure that the in phase lamp on the Rider Board is on (if fitted) and the OUT PHASE LED on the feedback board is not on.

E. Control Indicators

The Diagnostic Board located on the control panel has a number of LED's (Light Emitting Diodes) to reflect the controlling mode of operation. These may be used as an aid in the initial load set-up. It is normal for the power, inverter voltage, TOT, and ramp LED's to be energized with control power on (breaker closed) but heat off.
(Push RESET button to extinguish trip LED's)
For systems with a factory ordered operator's panel:
If the water and cabinet temperature are within specified limits and the water pressure proper, the RESET button can be pushed and the high frequency power ready lamp will energize on the control panel. Where the customer has made use of the "customer interlocks provision", some conditioning may be required at the load to get a ready lamp.

F. Starting Power Supply

Before asking for heat, turn the control dial to the lowest setting. This will allow the unit to be energized at a preset minimum low power level. Ask for heat from the power supply by pushing the start switch or other related start circuitry (user altered).

The HEAT ON lamp should energize.

Turning the control dial clockwise will cause the voltmeter and the kilowatt meter to increase. The frequency meter will show the inverter operating frequency dependent on the load. The power supply incorporates preset limits for protection of the equipment and load with regards to frequency and voltage. However, if any transformers are used between the power supply output and load, they should be tapped to remain within the voltage range of the load capacitors.

G. Limit Circuits

When operating into an unproven load, various limits may be reached before rated power is obtained. Operating on a limit is not detrimental to the power supply, and it is not abnormal to obtain a limit condition when operating at rated output of the power supply.

CHAPTER 2 OPERATING INSTRUCTIONS

Section I. Operator's Control Indicators

A. Location

All controls are conveniently located on the operator's control panel with lights, meters and controls provided. The operator's panel may be located at the power supply or ordered as a separate panel to be mounted in a remote enclosure. Large indicating labels, colored lamps and a large control dial are provided for the operator's convenience.

B. Function and Meaning

1. **Meters – CAUTION: THE METERS HAVE BEEN FACTORY SET AND SHOULD NEVER BE ADJUSTED OR TAMPERED WITH.**
   Kilowatt (Zero to 160% reading) – 100% indicates power supply is delivering rated power to load. Near optimum operating conditions should exist at the load.
   Voltmeter (Zero to 160% reading) – 100% indicates power supply is operating at maximum rated voltage. If rated kilowatt has not been obtained and power supply is running with limit lamp energized and 100% voltage, load matching is required.
   Frequency (Zero to 160% reading) – 100% indicates power supply is operating at normal frequency. The load resonant frequency is slightly below this.
   Ground Leakage (Zero to 160% reading) – 100% indicates 4A of current leakage to ground from the power supply and load circuit. This can indicate onset of a ground fault.
2. **Lamps**
   Lamps are provided to indicate a condition of normal operation, a condition of limit and indication of fault or shutdown. The following is provided as a brief description:
   Power ready – Provides an indication that the power supply is ready to provide power to the load on demand. For the ready lamp to energize, the breaker must be on, adequate water pressure and temperature must exist and customer interlocks (where used) must be energized. The lamp may not energize if any of the previous conditions are not satisfied. (Used only on units manufactured before 1987).
   Heat On (Inverter On) – Indicates the power supply has been energized to the point of producing power. The kilowatt and voltage meters should indicate some level of output. The frequency meter should indicate near the load resonant frequency.
   Fault – Indicates that the power supply has been stopped for protection of the power supply or load. When this lamp lights, the diagnostic board will indicate the specific reason for the fault.
   Water Hi Temperature – Indicates that the maximum allowable inlet cooling water for the power supply has been exceeded. (Maximum of 46° C, 115° F allowed). Check water temperature. Used on units manufactured before 1987; LED indicators were used from 1987 onwards. This

interlock also occurs when any inverter return exceeds 150° F, or furnace return exceeds 185° F.

Water Low PSI – Indicates that the water pressure has fallen below the minimum requirement of 30 pounds per square inch. This is monitoring the difference between inlet and outlet pressure. Where site drains are used, no back pressure exists at the outlet, so the inlet may be near 30 PSI before stopping the power supply. However, where a closed loop system exist with some distance used in the return, considerable backpressure may exist and the inlet pressure must be high enough to maintain a 30 PSI difference between the two. Water hammer should be avoided in either case to avoid nuisance stopping. (Used on units manufactured before 1987; LED indicators were used from 1987 onwards.) This interlock is also used to indicate faulty positioning of furnace selector switches (where used).

Voltage Limit (LED) – Indicates that the power supply is operating on the inverter voltage limit. If more power output is desired, increase amount of charge material.

Current Limit (LED) – Indicates that the power supply is operating on the AC current limit. This is a normal condition with a cold charge.

Power Limit (LED) – Indicates that the desired power level, as requested by the control dial, has been attained.

Frequency Limit (LED) – Indicates that the output frequency should be reduced. (Normally add capacitance.)

TOT Limit (LED) – Indicates that the preset level of turn-off-time on the inverter SCR's has been reached. This is a normal running condition.

Ramp Limit (LED) – Indicates firing frequency is at its upper limit.

Rectifier ON (LED) – Indicates that the rectifier is being fired. (No longer used after 1987).

Inverter ON (LED) – Indicates that the inverters SCR's are being fired. (No longer used after 1987).

Rectifier Full On (LED) – Indicates that the rectifier output voltage is at its maximum level. (No longer used after 1987).

Voltage Trip (LED) – Indicates unit has shutdown in response to excessive voltage across an inverter SCR.

Current Trip (LED) – Indicates that the input current has been excessively high and has stopped the unit.

Ground Trip (LED) – Indicates that a ground fault has caused more than 4A of current to flow and has stopped the unit.

***3.* Operator's Controls**

Start Switch – Allows for starting the power supply by pushing the switch. This requires a "High Frequency Power Ready Lamp" before the operator will have control.

Stop Switch – This switch performs two functions. It allows for stopping the power supply at the operator's control panel. It also allows for RESET of the power supply if the trip lamp is energized.

Control Dial – Provides the operator a means of changing the desired level of power. (May be controlling voltage in some setups.) The dial allows for obtaining minimum levels of power when in a counter-clockwise position. Rotation in a clockwise direction will automatically increase the power to the level that the operator desires.

Leak Detector Test – Will cause an indication of leakage current when pressed and can be used to test the ground trip function.

Section II. Controls, Features, Remote Interlocks

The equipment includes a number of control features and interlocks included for personnel and equipment safety. The following is provided as a brief description of these. The control schematic of the customer drawing set may be referred to for a reference.

A. 24 Volt DC Battery Operated Supply

Provides low voltage control power for preliminary energization and protective circuitry of the power supply. It is located within the control cabinet.

B. Breaker Ready

Provides an indication that the breaker is ready to be closed. For the lamp to energize, all doors must be closed and secured with the enable switch for the 24V DC. Multiple cabinet systems require that all doors be closed on each section to energize the lamp. The enable switch must be held while closing the breaker handle.

CAUTION: NO ENTRANCE TO THE POWER SUPPLY SHOULD BE ATTEMPTED WITH THE BREAKER CLOSED.

Any attempt to open any door of the power supply will drop out the breaker through interlocking and require having to reset each breaker utilized. A provision is made for the user to connect his load station doors into this circuit also, so when used, these doors must also be closed for the breaker ready lamp to energize.

C. Control Interlocks for the User

Provisions for the user to break the 24V breaker holding circuit, the water pressure, water temperature and START/STOP circuitry are provided. Refer to the control schematics and wiring diagrams provided with the equipment.

D. Protective Switches

The following sensors are used to protect the equipment. Corrective action is also listed.

1. Water Temperature

   Inlet water temperature is high.

   Corrective Action: Lower water temperature. This switch will recycle within a 10° F drop in water temperature.

   NOTE: On closed loop systems, this lamp is also controlled by circuits within the power supply monitoring individual water lines. Each line monitor has a lamp to indicate the high temperature line. The lamps receive power from the 24V DC supply. With the breaker off and doors open, a small toggle switch on the 24V-battery board in the control cabinet can be switched to supply 24V DC to the lamps for the monitors. A

faulted line must have its sensor manually reset before again starting. A small RESET switch is located on the sensor.
Corrective Action: Check the faulted line for kinks in the hose or gradual deposits in the water line that restrict flow. The toggle switch must be switched to the breaker position before the breaker can be energized.

2. Pressure Switch
Water pressure is low.
Corrective Action: Raise inlet pressure.
NOTE: Where closed loop systems are used, too much back pressure may exist or a valve is closed in the water system.

E. **Start Switch**
Allows for starting and obtaining medium frequency power from the power supply.

F. **Stop/Reset Switch**
Allows for stopping the medium frequency power of the power supply. Also used to reset the control.

G. **Control Dial**
Provides a means of changing the level of power delivered by the power supply. (This dial will provide a means of changing voltage when the equipment is conditioned for a voltage mode).

Section III. Sequence of Operations

The following is provided as a start sequence procedure for the power supply. It is intended for the unfamiliar operator of the equipment:

A. **Energize the cooling water system for the power supply.**

B. **Verify that pushing the enable switch energizes the "Breaker Ready" lamp located near the breaker handle of the master cabinet.** (Each cabinet for a multiple cabinet system). If not, check all doors for being closed properly, or load station doors if connected to power supply. Ensure external "input" line voltage is applied to the cabinet. Never try to engage the circuit breaker without pushing breaker enable button. This reduces the life of the breaker. **CAUTION: Before closing breaker, follow all safety precautions at load. Dangerous voltages exist at the work coil.**

C. **Push the enable switch and while holding it in, close the breaker.**

D. **Proceed to the Operator Panel. Push the STOP/RESET button. Verify that the Power Ready lamp is lit.** If not, check to insure all fault lamps are out. If no fault lamps are lit and the power ready lamp is not on, check the load equipment where interlocking has been incorporated into the start circuit of the power supply. Rotate the control dial counter-clockwise.

E. **Push the start switch for heat. Generally, the work coil would be loaded at this time.** Some systems may be set up for automatic cycling by the user.

F. **Rotate the control dial on the operator's panel for the desired output of the power supply.** If desired levels of rated power or voltage cannot be reached, the load matching may be required.

G. **To stop the unit, push the STOP switch on the operator's panel.** The unit may be again started with the control dial at the desired operating range as previously used.

H. **CAUTION: BEFORE WORKING AROUND THE WORK COIL OR CAPACITORS OF THE LOAD, ALWAYS TURN BREAKER(S) OF POWER SUPPLY(IES) OFF FOR SAFETY.**

CHAPTER 3 CIRCUIT DESCRIPTION

Section I. Power Section

A. Basic Components

The Pillar MK8 converter consists of a variable voltage, variable frequency AC source connected to an induction heating or melting load coil (refer to Figure 1). A slightly more detailed diagram in Figure 2 illustrates the salient components in the unit. (i.e. rectifier filter, inverter and the load).

1. Rectifier

The rectifier consists of six (6) SCRs and a free wheeling diode. In simple terms, it converts the three phase input power into a DC supply which is variable in voltage output. The presence of the free wheeling diode restricts the output voltage to only positive values, increasing the efficiency of rectification and providing a current path for the DC choke current when the SCRs are turned off.

2. Filter

The filter features a DC choke and a smoothing capacitor which are used to reduce the amount of ripple on the DC bus.

3. Inverter

An SCR and a series reactor form the inverter section and are used to transmit pulses of DC energy from the DC bus to the load circuit.

4. Load

The load consists of the tank capacitors (which are usually contained in the cabinet), the heating or melting coil and the load resistance which represents the material to be heated within the coil.

B. Principles of Operation

The MK8 can be reduced to the form as shown in Figure 3 to simplify explanation. In Figure 3, we have a fixed DC supply, which is equivalent to the rectifier when producing a steady output. The SCR is pulsed every cycle of tank oscillation to transmit a pulse of current into the load. Refer to Figure 4. Typically, the SCR must be turned on at some point in the cycle such as “t1”. The instantaneous polarity of current then flows through the SCR and reactor Ls, providing power to the tank circuit. As the instantaneous voltage of the tank rises, a peak current is attained, following by a reduction to zero. This occurs at time “t2” and extinguishes the SCR. At time “t3” the cycle repeats. To increase the energy transmitted to the tank, there are two possibilities:

a. Advance the firing point from “t1” to increase the source voltage that drives current into the load, thus producing a more energetic pulse.
b. Increase the DC potential, E, which similarly to (1), increases the SCR forward potential, at firing. Increasing the DC potential, E is not possible with a fixed DC supply as shown in Figure 3, but is easily achieved with a controlled rectifier as shown in Figure 2. In fact, both techniques are used to control power. At low power levels, the firing point of the SCR is held at a fixed angle with respect to the tank voltage. During this zone of operation, varying the output of the rectifier controls power variations. This could typically be the case up to 30% power.

Above this level, the rectifier is producing maximum voltage output, and thus, to increase power further, the firing angle of the SCR must be advanced. To equalize the DC potential with respect to ground and to also maximize possible tank voltage, two inverter SCRs are used as shown in Figure 5. The basic power sizes of the MK8 as shown are 150 kW at 600 Hz or 1200 Hz and 100 kW at 3000 Hz. Where higher power levels are required, extra inverter SCRs are added in parallel. Refer to Figure 6.

Section II. Control

A. Interlocking and Safety Devices

CAUTION: LETHAL VOLTAGES PRESENT. DO NOT DEFEAT ANY INTERLOCKS. WHENEVER WORKING INSIDE CABINETS, DISCONNECT ALL INPUT POWER.

1. **Circuit Breaker Interlock**
   Door switches are provided in series with a 24V DC holding coil in the circuit breaker. Terminals for additional customer interlocks are provided; refer to schematic and wiring diagram. A 24V DC Battery Board provides the 24V to the breaker. In order to turn on the breaker, it is necessary to first depress a button (PB3) which connects the batteries to the CB holding coil (providing door switches and interlocks are closed). After turning the break on, a rectifier 24V Battery Board supplies 24V DC to the circuit breaker.
2. **Grounding Contactor**
   A grounding contactor (GK) will short the output bus to the cabinet whenever the circuit breaker is on, but unit is not running. Upon pushing the start button, the contactor will open and allow the power supply to start.
   WARNING: CABINET(S) MUST BE PROPERLY GROUNDED OR LETHAL ELECTRICAL SHOCK MAY RESULT.

3. **Water Pressure and Temperature**
   All temperature and pressure monitoring circuits incorporate "Fault Memory". The RESET button must be pushed to clear the fault. This memory circuit aids in diagnosing momentary pressure or temperature fluctuations.
   Protection provided:
   i. Differential pressure switch (30 PSI). Gauges are provided to monitor inlet and outlet pressure. On units with internal water systems, two sets of gauges are provided; one set of gauges for external pressure and one set of gauges for internal pressure.
   ii. Inlet water temperature switch 115° F (46° C).
   iii. Individual outlet temperature switches, 150° F (65.5° C). A light directly across each outlet switch aids in detecting the overheating water circuit. The drain temperature switches must be manually reset.

In addition, melting units provided protection for:

iv. Furnace outlet drain switches 185° F (85° C).

v. Furnace pressure switch (30 PSI).

4. **Other**
   Provisions are incorporated for additional interlocks. Refer to Power and Control schematic for details. Other interlocks or control functions may be present.

B. Electronic Control System (units produced before 1987)

1. **START/STOP Board** provides signal for starting, stopping, over voltage tripping, over current tripping and ground leakage shut down.
2. **Regulator Board** controls the power level output of the unit. It responds to feedback signals of current, voltage and output frequency to hold the unit within limit constraints of these variables. It also generates the firing pulses for the rectifier SCRs in response to desired power level and input voltage phase information.
3. **Inverter Firing Board** controls the firing of the inverter SCRs. By sensing the timing of the output voltage and turn-off-time (TOT) of the SCRs, the board produces suitable firing pulses.
4. **SCR and Bus Monitor Board** provides feedback information representing SCR voltage and DC bus voltage. It also produces TOT information derived from the SCR voltages.
5. **Leak Detector Board** senses any ground current. It provides an indication of any current and stops the unit if it exceeds 4A.
6. **"B" Supply** assembly provided +/-20V unregulated power to the electronic control boards.

C. Electronic Control System (units produced during 1987 and thereafter)

The MK8 has an integrated control and operator's panel, which has two electronic control circuit boards.

1. **Feedback and Amplifier Board**
   This board couples the main control board to the power supply. All the feedback signals enter this board and are reduced to an acceptable level before entering the main control board. The firing signals generated by the main control board are amplified by this board before driving the SCR's. This board also contains the +20V and –20V unregulated supplies and an indicator if phase rotation is incorrect.
2. **Main Control Board**
   This board contains all the circuitry to start/stop, regulate and monitor the MK8 Power Supply. It also contains the lamp/meter drivers and temperature control interface circuitry.

D. Other Control Assemblies

1. **Battery Board**
   This board contains low voltage battery cells for powering the circuit breaker release coil when first energizing the unit.
2. **Rider Assembly**

The rider assembly monitors and steps down the incoming three phase voltages. Also incorporated, on pre-1999 units, are two lamps, which indicate when the three phase sequence is correct for initial installation.

E. Control Block Diagram

Figure 7A and 7B are block diagrams showing the important interconnections from the power circuit to the electronic controls. Figure 7A refers to units produced before 1987 and Figure 7B refers to units produced during 1987 and later.

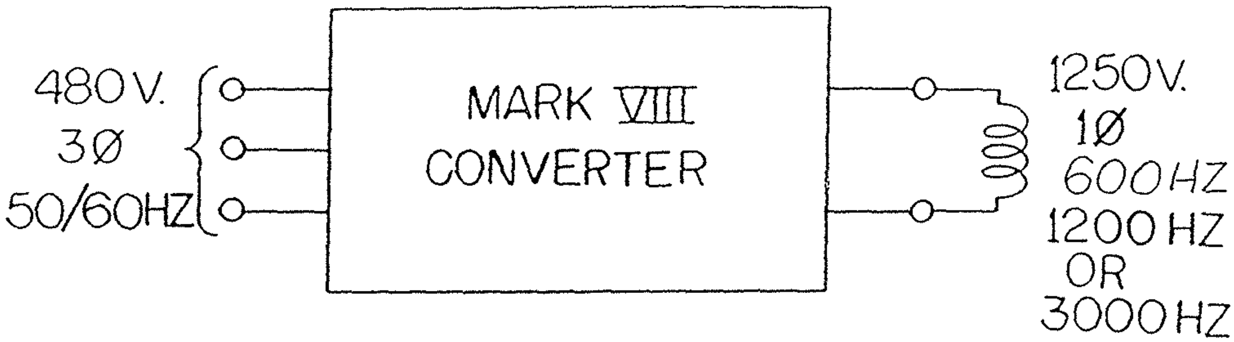
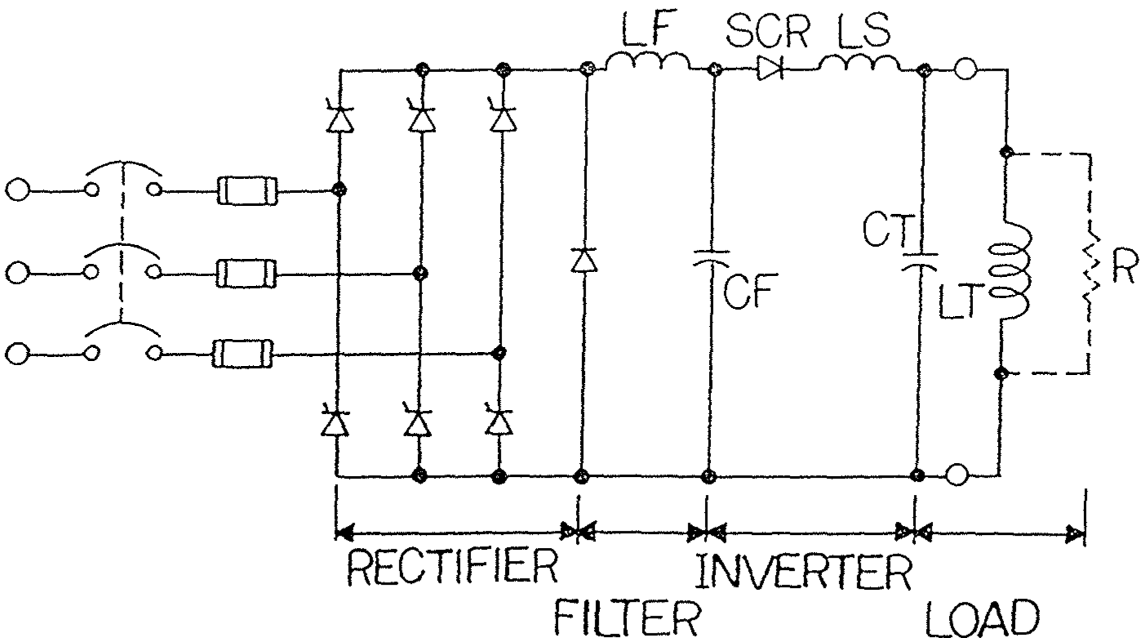


FIGURE 1

FIGURE 2



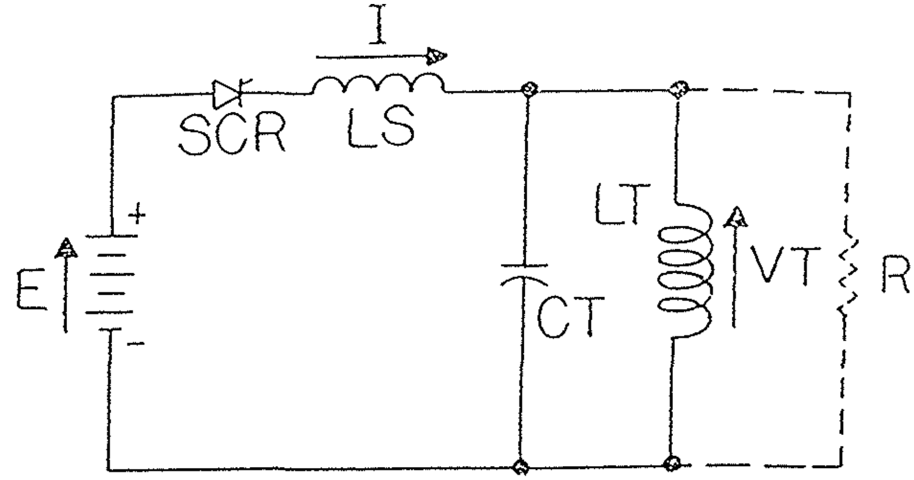


FIGURE 3

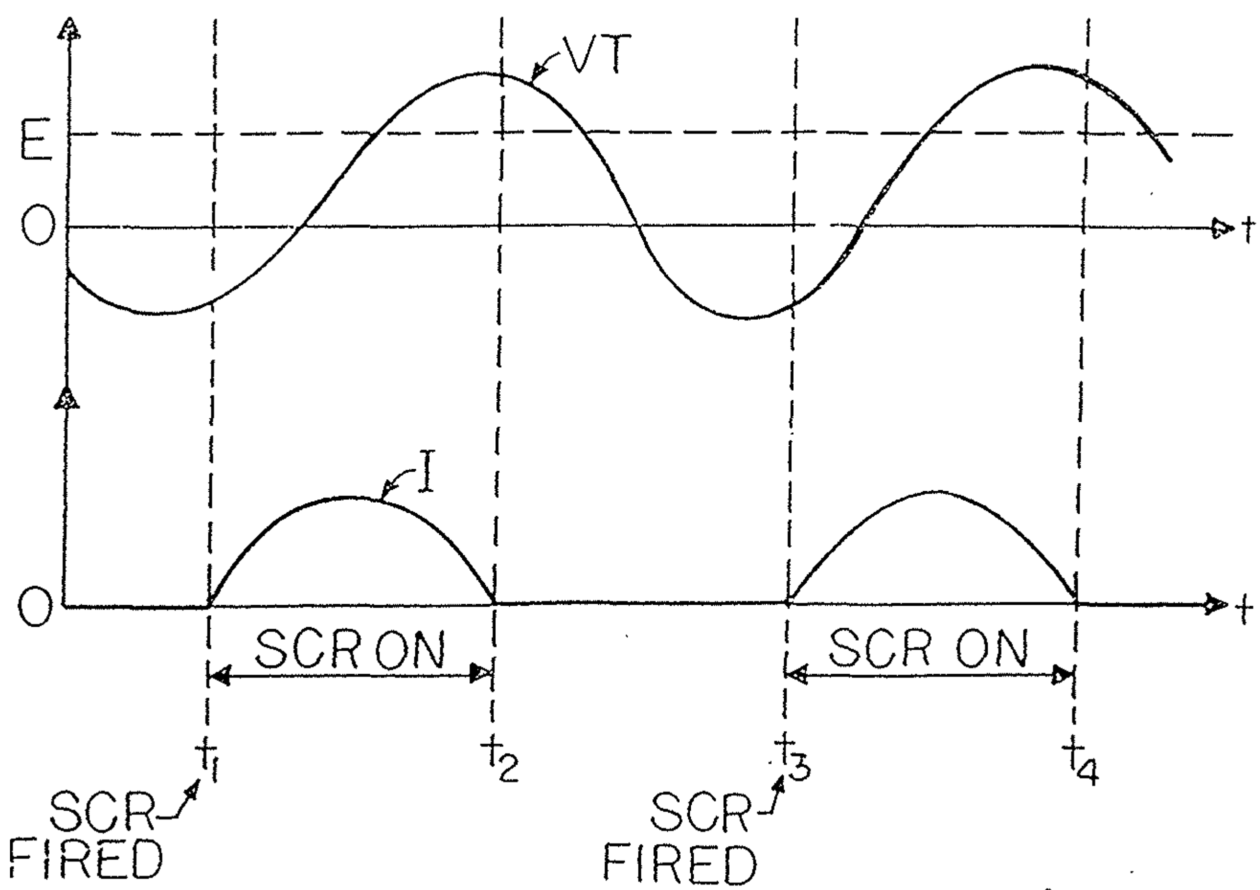


FIGURE 4

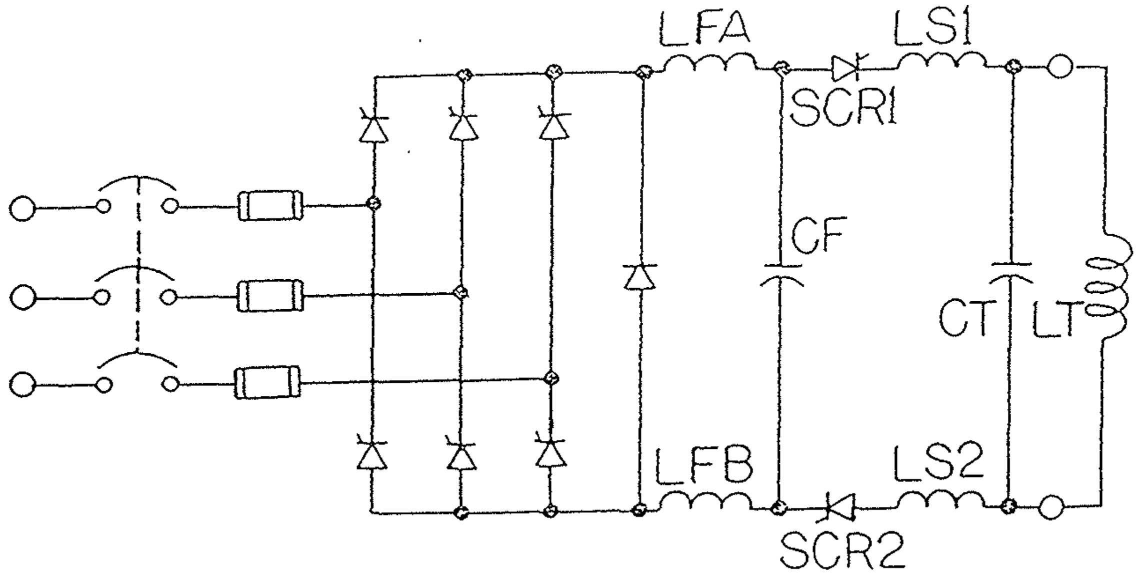


FIGURE 5

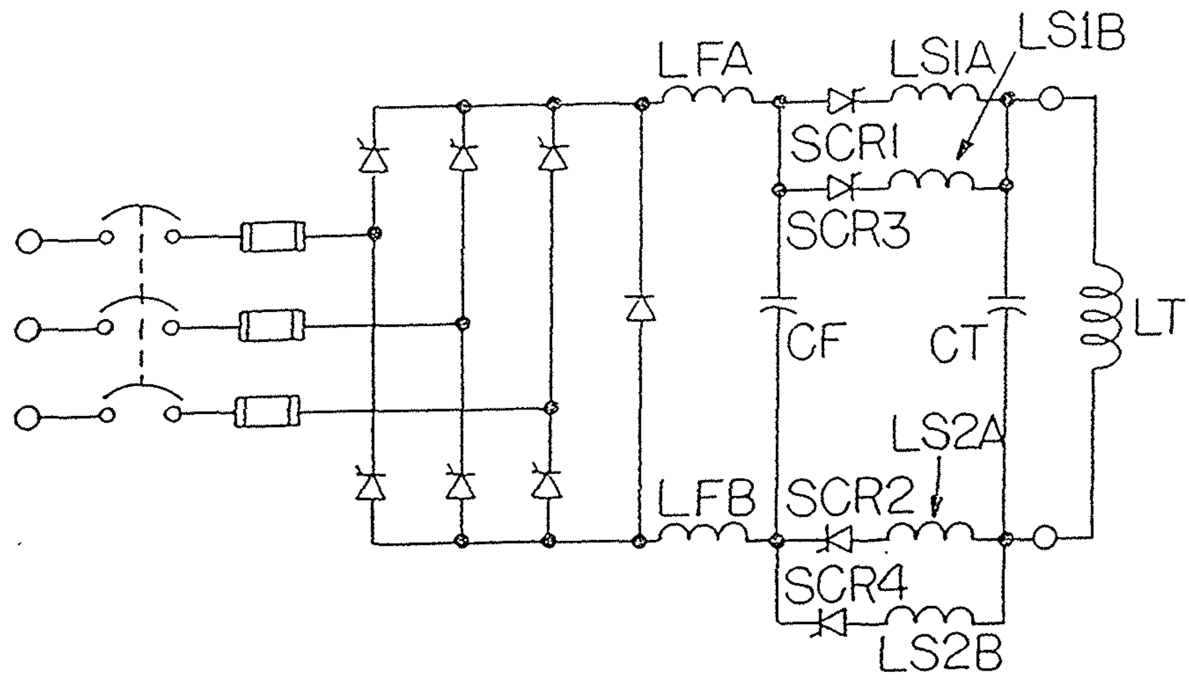


FIGURE 6

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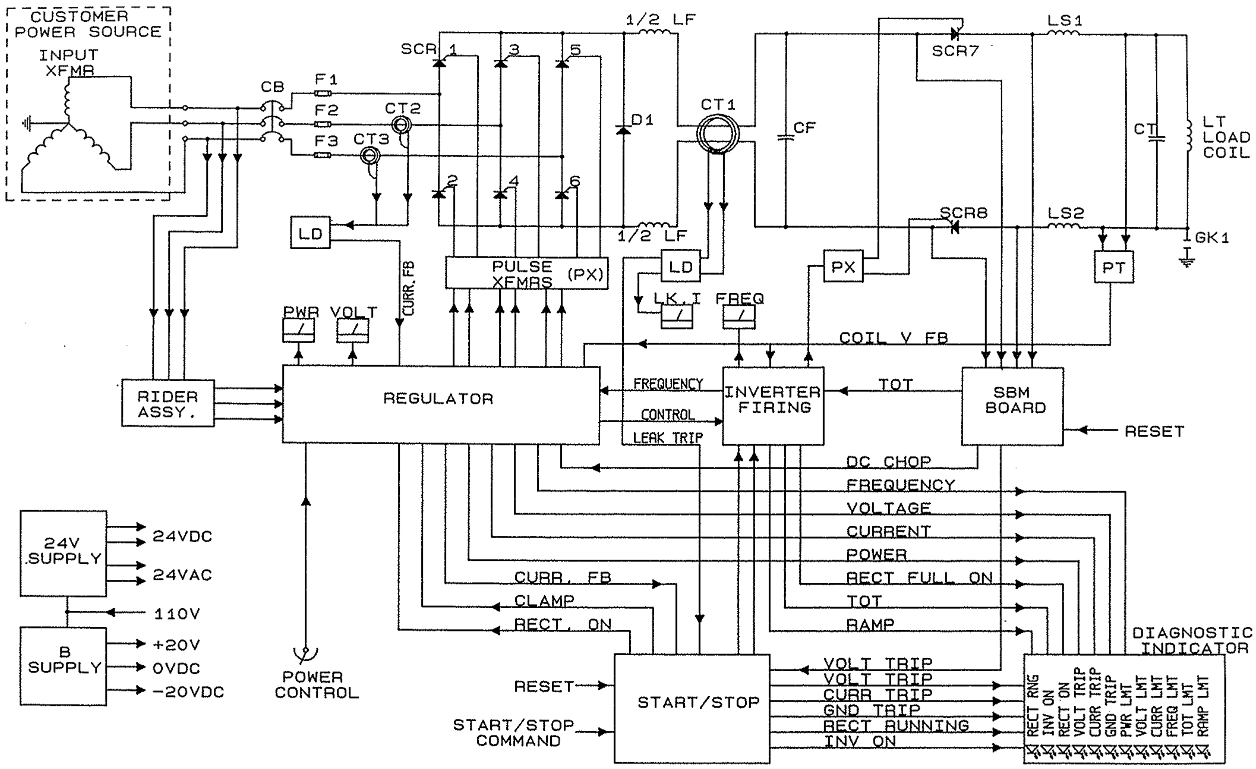

Figure 7A

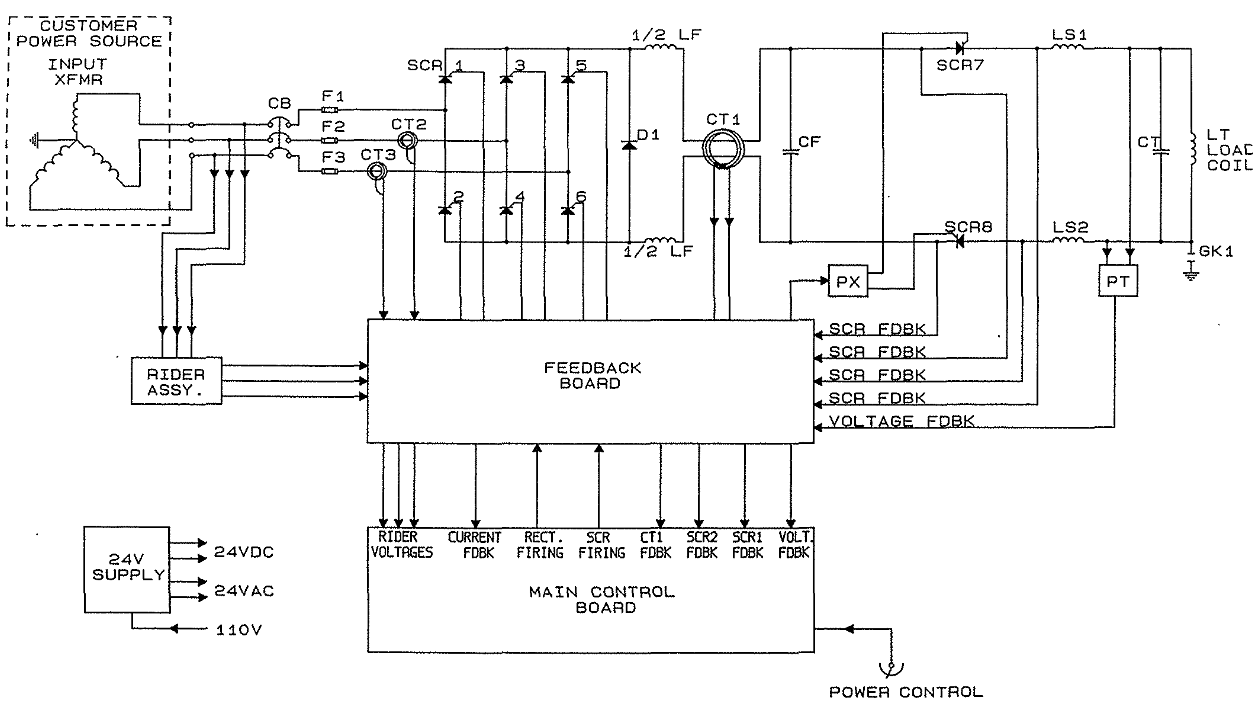

Figure 7B

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CHAPTER 4 MAINTENANCE

Section I. Safety Precaution

A. NOTES – CAUTIONS – WARNINGS

1. Any equipment of this type should be worked on with caution, particularly when the power supply is operated with the cabinet doors open. All standard safety procedures should be followed. <u>NEVER TAKE CHANCES!</u>
2. <u>Always wear protective eyeglasses while working within the cabinet.</u>
3. <u>Do not stand in water or on grounded surfaces while reaching into the cabinet.</u>
4. Capacitors store charge! Never trust a capacitor to bleed off completely. A meter or ground strap should be used to check each stud before handling. Some capacitor studs, when not tied to buswork (not used), may build up a considerable static charge. Ground before handling.
5. Never work alone on equipment. Always have other personnel present when working with potentially lethal voltage.
6. Trained personnel in electrical shock and/or cardio-pulmonary resuscitation should be available. For CPR training contact the American Red Cross or the American Heart Association.

B. First Aid for Electrical Shock

The following is provided as a reference in the event of electrical shock to personnel. It is intended for those who have no knowledge of what to do. Local agencies should always be consulted if any such accident occurs.

1. Rescue

In case of electrical shock, shut off voltage at once and ground the circuits. If the high voltage cannot be shut off without delay, free the victim from contact with the live conductor as promptly as possible. Avoid direct contact with either the live conductor or the victim's body. Use a dry board, dry clothing, or other non-conductor to free the victim.

2. Symptoms

a. Breathing stops abruptly in electric shock if the current passes through the breathing center at the base of the brain. If the shock has not been too severe, the breath recovers after awhile, and normal breathing is resumed, providing a sufficient supply of air has been furnished meanwhile by artificial respiration.

b. The victim is usually very white or blue. The pulse is very weak or entirely absent and unconsciousness is complete. Burns are usually present. The victim's body may become rigid or stiff in a very few minutes. This condition is due to the action of electricity and is not considered rigor mortis. Artificial respiration must still be given as several such cases are reported to have recovered. The ordinary and general tests for death should never be accepted.

3. **Mouth-to-Mouth Artificial Respiration**
   Start artificial respiration immediately. Do not wait for a mechanical resuscitator, but when an approved model is available, use it. At the same time, send for medical help if assistance is available. **Do not leave the victim unattended.** Perform artificial respiration at the scene of the accident, unless the victim or operator's lives are endangered from such action.
   In this case only, remove the victim to another location, but no further than necessary for safety. If the new location is more than a few feet away, artificial respiration should be given while victim is being moved. Artificial respiration, once started, must be continued without loss of rhythm. The mouth-to-mouth artificial respiration is described here.

   a. Position of victim. Place victim in the face upward position and kneel close to his left ear.
   b. Clear the throat. Turn the head to one side and quickly wipe out any fluid, mucus or foreign body from mouth and throat with the fingers.
   c. Open and align air passages. Tilt the head back and extend the neck to open and align the air passages so that kinking or pressure does not block them.
   d. Lift jaw forward. Place thumb into the mouth and grasp the jaw firmly. Lift the jaw forward to pull the tongue forward out of the air passages. Do not attempt to hold or depress tongue.
   e. Pinch nostrils closed. Use other hand to keep victim's nostrils pinched closed to prevent air leak.
   f. Form tight seal with lips. Rescuer's wide-open mouth completely surrounds and seals the open mouth of the victim. This is not a kissing or puckered position - the mouth of the rescuer must be wide open.
   g. Blow. Exhale firmly into victim's mouth until the chest is seen to lift. This can be seen by rescuer without difficulty.
   h. Remove mouth and breathe in. During this time, rescuer can hear and feel the escape of air from the victim's lungs.
   i. Repeat steps f, g and h. **Continue at a rate of 12 to 20 times per minute**.

   CAUTION: Excessively deep and rapid breathing by the rescuer may cause him/her to become faint, to tingle or even lose consciousness. Breathing should be normal in rate with only moderate increase in volume. In this way, rescue breathing can be continued for long periods without fatigue.
   NOTES: Keep airway clear of fluid and other obstruction. Re-adjust position if air does not flow freely in and out of victim. Keep neck extended and chin pulled forward.

C. Methods of Interlocking and Defeaters

The power supply has a number of interlocks provided for personnel and equipment safety. The following is a brief description of these. It is intended to familiarize the person servicing the power supply with its interlocking features.
CAUTION: Lethal voltages present inside equipment. Under no circumstances should unqualified personnel be allowed to work in equipment.

1. Door Interlocks

All doors electrically interlocked to the breaker mechanism using the 24V DC supply. Opening any one door of a system will drop out the breaker disconnect or all breakers for multiple cabinet systems. However, this should not be done intentionally. The useful life of these mechanisms are limited. The power supply may be operated with the doors open if the door interlocks are manually operated. Only qualified personnel should be allowed to operate the power supply this way, and then only for maintenance or service purposes. Pulling out on the door interlock mechanism of all doors opened will override the interlock. Provisions are made for the load station to be interlocked with the breaker ready. Thus, where used, may prevent obtaining a breaker ready lamp if the load station doors are open or not in an override position.

2. Power Ready Interlock

To obtain a ready light, proper water pressure, water and cabinet temperature, and any user connected interlocks (provision for connections provided) must exist. In closed loop systems, each water line within the power supply is monitored for excessive outlet temperature resulting from inadequate or restricted flow. These temperature switches require a manual reset once energized. Where several are used, the lamp will define which of the lines have run hot. Each of these circuits is indirectly tied to the high water temperature lamp on the operator’s panel. Thus, if the temperature lamp energizes and cooling the inlet water down does not de-energize the lamp, it is probable that an internal temperature switch is energized. Power may then be removed from the cabinet and the doors may then be opened. With the circuit breaker off, and the doors open, a small toggle switch on the 24V battery board can be switched to supply 24V DC to the lamps on the drain temperature switches.

Section II. Preventive Maintenance

The Pillar Power Supplies are designed to survive the rugged use that may be imposed upon them under rough environments. However, preventive maintenance can become an important aspect with respect to long life and durability for equipment of this nature. The following is provided as a guide in periodic preventive maintenance checks.

A. Visual Inspection

A periodic visual inspection is good practice. One should check for leaks, dust build-up or evidence of condensation, etc. that is detrimental to the equipment. Heavy build-up of foreign matter, particularly where metallic particles exist, may reduce the level of voltage that normally is supported by devices and buswork. An air hose if available, is convenient for removing dust from the equipment. Minimum build-up should occur if the power supply doors are kept closed except for maintenance purposes.

B. Targets

Targets are used in the water circuits to prevent degradation of the copper tubing via electrolysis where DC potentials exist. Without these targets, erosion of tubing is probable dependent on the existing water conditions. The targets are always located at the positive end of the water path between copper tubes where any DC potential exists. It is the low potential side of the tubing in the water circuit that will build up deposits regardless of the water flow direction. It is good practice to check conditions of the lines after the equipment has been in operation for some time. The quality of water at the site will determine the severity of corrosion. See Item 6 of Cooling Water Requirements in Chapter 1 for recommendations for Quality of Water.

C. Condensation

Condensation of water on the water-cooled tubing should be avoided. This will occur if very cold water is fed through the system; for example, well water. The accumulation of water may reduce the margin of voltage that can be supported in the cabinet enclosure or provide a high impedance path to the frame ground. Where this condition exists, higher water temperatures of the water lines (cannot exceed 115° F for continuous operation) or removal of moisture from the cabinet air are recommended.

D. Water Flow

Adequate water flow through the water lines is necessary for proper operation. The equipment monitors water temperature and pressure relative to proper flow. Where closed loop systems are employed, temperature sensors are utilized to monitor each outlet line as a means of guarding against gradual restriction of water in their respective lines. With the circuit breaker off, a small toggle switch on the 24V battery board can be switched to supply 24V DC to the lights across each outlet temperature sensor. This will indicate which drain has overheated. The operator should periodically monitor the water lines where sight drains are used to provide a visual inspection of water flow.

E. Connections

Power and bus connections should remain tight as sent from the factory or after being secured. However, carelessness or thermal contraction and expansion could result in an improper connection. A resistive joint may thus occur. This is usually recognized by discoloration of the buswork, inability to get rated power where it had been obtained previously or unstable operation of the equipment. During periods of diagnosis for unusual troubles or annual shutdown maintenance, it is good to check connections. Over torque of connections should be avoided. Tables for torque specifications of critical components are provided in the next chapter. Terminations on control boards (even though rugged) should be handled with care so as not to over tighten and break the screw head.

Section III. Corrective Maintenance

The following information is provided as an aid to diagnosis. Reference material and formulation of techniques that may be used to trouble-shoot the power supply are provided. It is expected that people making these checks will have some previous familiarization with electrical circuits and equipment. However, many of the procedures herein may be readily used by persons not familiar with trouble-shooting techniques.

CAUTION: LETHAL HIGH VOLTAGES ARE PRESENT AND EXPOSED WITHIN ENCLOSURE WHEN THE BREAKER IS CLOSED. ALL SAFETY PRECAUTIONS SHOULD BE OBSERVED BY PERSONNEL WORKING INTERNALLY IN THE EQUIPMENT WITH POWER APPLIED.

A. How to use and Ohmmeter

A simple ohmmeter is a very useful instrument that may be used for the location of faulty components or for tracing the continuity of circuits. It allows one to make checks with no power on the equipment, thus, is the safest instrument for inexperienced personnel. Usually a VOM (volt-ohmmeter) is used for this purpose, as it requires no external power to operate the instrument, making it portable. A VOM of at least 20,000 ohms per volt should be used. Some meter manufacturers may design their meters to read polarity in the reverse direction and must be recognized for this fact when making polarity measurement (for example: diodes read in one direction only when good).

Semiconductors are to be checked on the R x 1 scale of an ohmmeter. Usually a selector switch is provided for selecting this range. With one end of the leads inserted into the meter (see meter instructions), and the other ends open, the meter will see an infinite resistance and remain at infinity. By touching the leads together, a short or zero ohms is seen by the meter. A control knob is provided on the meter for calibrating the meter to zero ohms. This must be adjusted before using the meter and checked for zero if moved to a different scale. If the meter cannot be zeroed, it is generally a sign that the battery is weak and needs replacing. See the following “Ohmmeter Tests for Semiconductors” and the troubleshooting tests section for other tests that may be performed with an ohmmeter.

As mentioned previously, the ohmmeter may be used to check continuity. The leads may be placed across a conductor and measure the resistance between the leads. A copper wire or bus will have a very low resistance and appear to be a short when looking across it. A resistance in the line between the leads will cause the meter to read the value if properly calibrated.

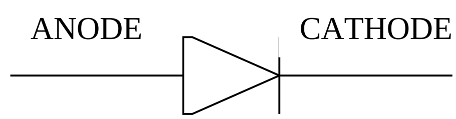
B. Ohmmeter Tests for Semiconductors

Set the multi-meter to ohms x 1 scale. The meter has an internal battery, which supplies power when used to measure resistance. Determine which lead of the meter is connected to the positive end of this battery. This can be done by checking a diode that is known to be good.

Semiconductors almost always fail short. Watch for sneak circuits when testing semiconductors in the circuit. Isolate suspected semiconductors and measure independently.

Diode

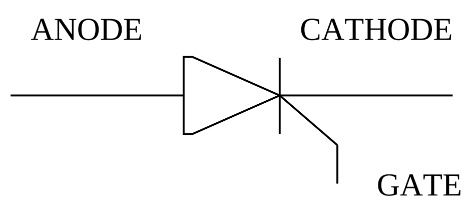
1. Place plus lead on anode and minus lead on cathode. Read approx. 30 ohms.
2. Place plus lead on cathode and minus lead on anode. Read >1000 ohms.



Note: This test is good for both small signal diodes as well as large power diodes.

Thyristor (SCR)

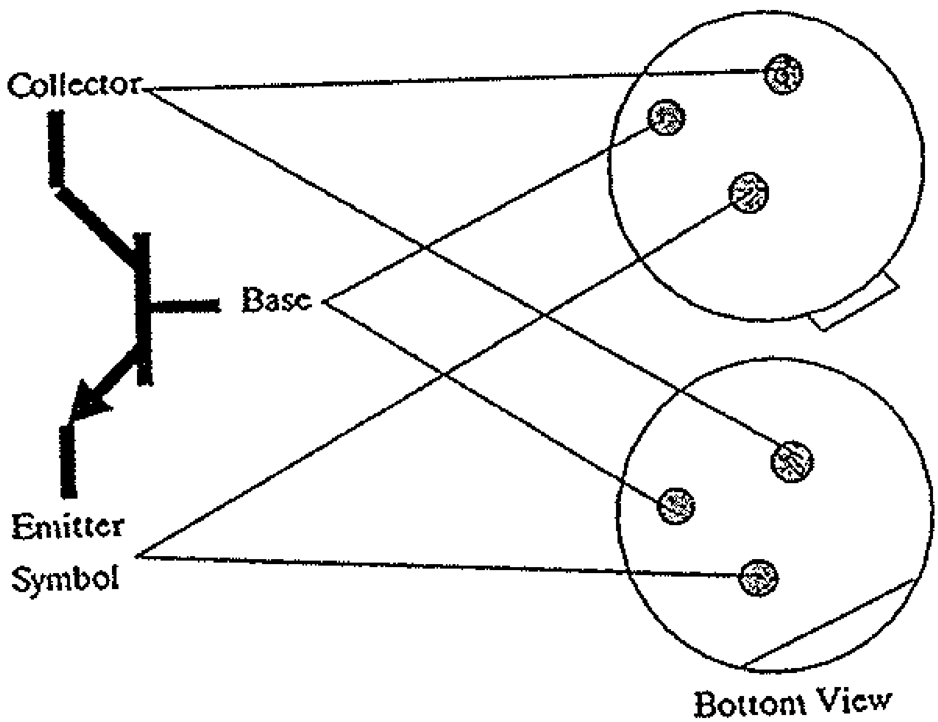
1. Place plus lead on anode and minus lead on cathode. Read >1000 ohms.
2. Place plus lead on cathode and minus lead on anode. Read >1000 ohms.
3. Place plus lead on gate and minus lead on cathode. Read approx. 30 ohms.



NPN Transistor

1. Place plus lead on base and minus lead on emitter. Read approx. 30 ohms.
2. Place plus lead on base and minus lead on collector. Read approx. 30 ohms.
3. Place minus lead on base and plus lead on emitter. Read >1000 ohms.
4. Place minus lead on base and plus lead on collector. Read >1000 ohms.
5. Place plus lead on collector and minus lead on emitter. Read >1000 ohms.

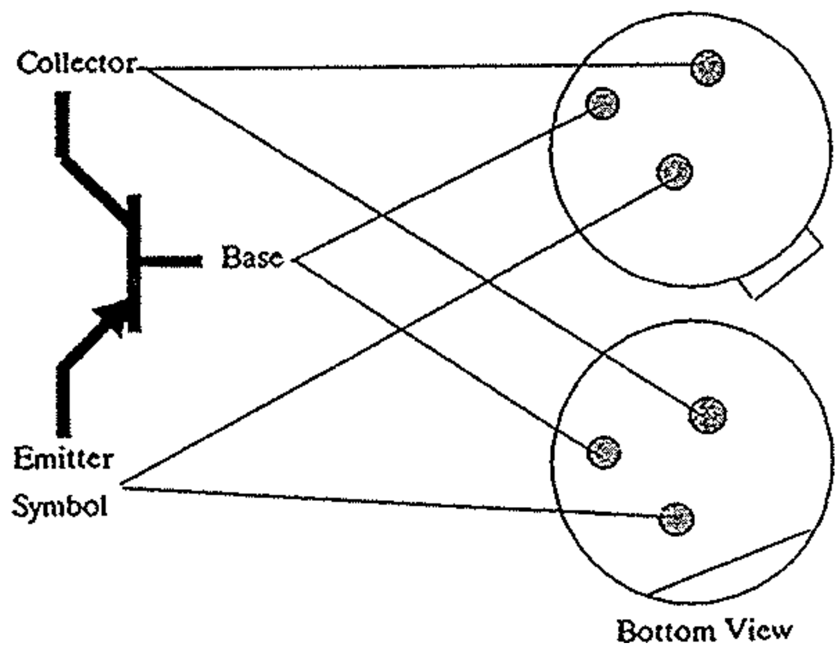
NPN Transistor



PNP Transistor

1. Place minus lead on base and plus lead on emitter. Read approximately 30 ohms.
2. Place minus lead on base and plus lead on collector. Read approximately 30 ohms.
3. Place plus lead on base and minus lead on emitter. Read more than 1000 ohms.
4. Place plus lead on base and minus lead on collector. Read more than 1000 ohms.
5. Place minus lead on collector and plus lead on emitter. Read more than 1000 ohms.

PNP Transistor



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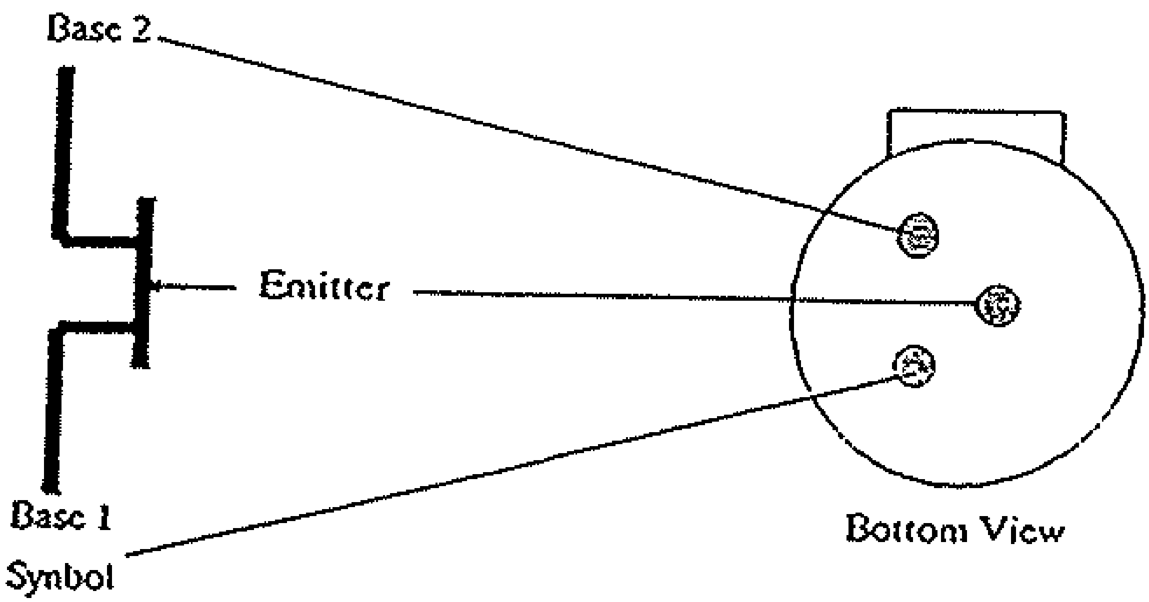
Unijunction Transistor

1. Place plus lead on emitter and minus lead on base 1. Read approximately 50 ohms.
2. Place plus lead on emitter and minus lead on base 2. Read approximately 50 ohms.
3. Place plus lead on base 1 and minus lead on base 2. Read approximately 1000 ohms.

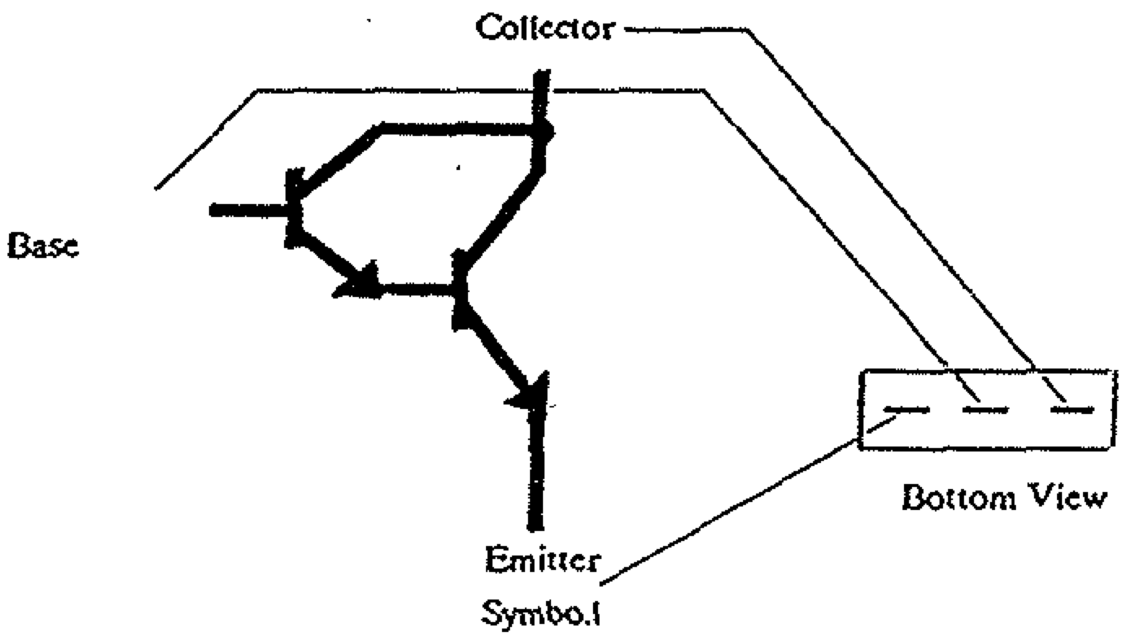
Unijunction Transistor



NPN Darlington Transistor

1. Place plus lead on base and minus lead on emitter. Read approximately 100 ohms.
2. Place plus lead on base and minus lead on emitter. Read approximately 15 ohms.
3. Place minus lead on base and plus lead on emitter. Read approximately 1000 ohms.
4. Place minus lead on base and plus lead on collector. Read more than 1000 ohms.

NPN Darlington Transistor



NOTE: Two transistors may be paralleled to forma a Darlington Transistor.

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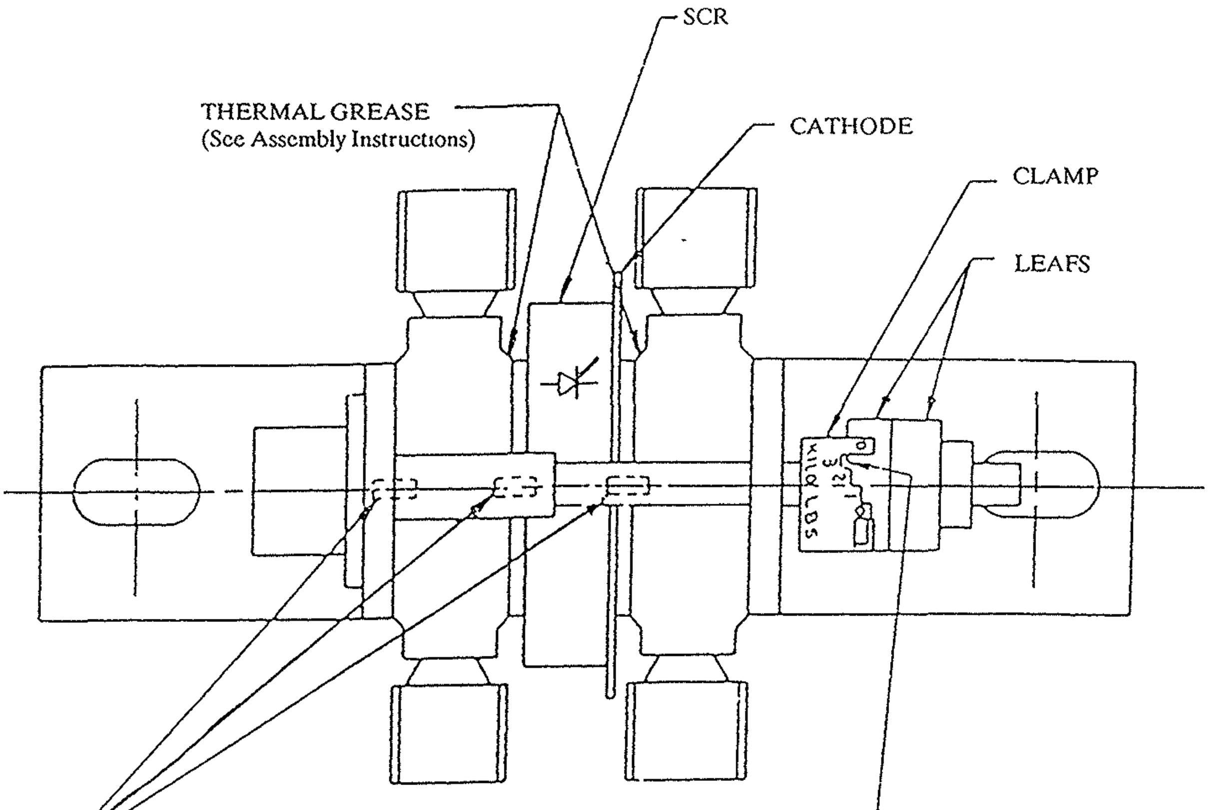
C. Procedure for Removal and Replacement of Power Semiconductors

1. Semiconductors

The following comments, picture and table are intended as a guide to be used in replacing any semiconductors. Care should be used in the handling and torquing of such devices, as well as the proper amount of grease. A convenient clamp and gauge are used on “flat pak” devices. The method of mounting and handling silicon rectifier cells is important for the following reasons. The cell can be damaged internally by rough handling or abuse. A low resistance electrical and thermal contact with the heat sink is necessary to prevent excessive heating of the cell. The original operation conditions must be maintained over a long period of time, even in a highly corrosive atmosphere.

a. Handling and Mounting Procedure for Flat Pak (Press Pak)
   i. Disc-type power semiconductor devices such as Press Pak were developed especially to dissipate power more efficiently using double-sided cooling. Because of the substantially lower thermal resistance achieved, the thermal resistance of the mating metallic surfaces becomes a significant portion of the total; consequently, unwanted variation in surfaces through neglect can cause excessive junction temperatures and ultimate cell failure.
   ii. Care must be used in handling and mounting these devices to avoid causing nicks, scratches and other deformations which effectively reduce the mating areas.
   iii. The heat dissipaters for double-sided cooling must be securely clamped parallel to the mounting surfaces conforming to instructions which include surface condition, thermal lubricant and clamping force. In general, the contact surface of each heat dissipater must be flat with a surface finish equal to the mating surface on the Press Pak. A self-leveling type mounting clamp assures parallelism and even distribution of pressure on each contact area. (See Figures 1 and 2 at the end of this section for assembly procedures.)

FLAT PAK ASSEMBLY (Inverter)



ROLL PINS, CHECK THAT ROLL PINS ARE PROPERLY MATED, 3 PLACES BEFORE TIGHTENING CLAMP

FIGURE 2

EDGE OF CLAMP TO BE ALIGNED WITH 2.5 NOTE ALTER TIGHTENING OF NUTS TO MAINTAIN EQUAL DISTANCE ON THREADS PRIOR TO TORQUING DEVICE, CHECK ZERO SETTING OF GAUGE WITH NUTS,

NOTE:

1. WITH TWO LEAFS AS SHOWN, FORCE = 2 x GAUGE READING = 5000 LBS
2. ON DEVICES WITHOUT A GAUGE, THE WHOLE ASSEMBLY MUST BE REPLACED.

FLAT PAK ASSEMBLY (Rectifier)

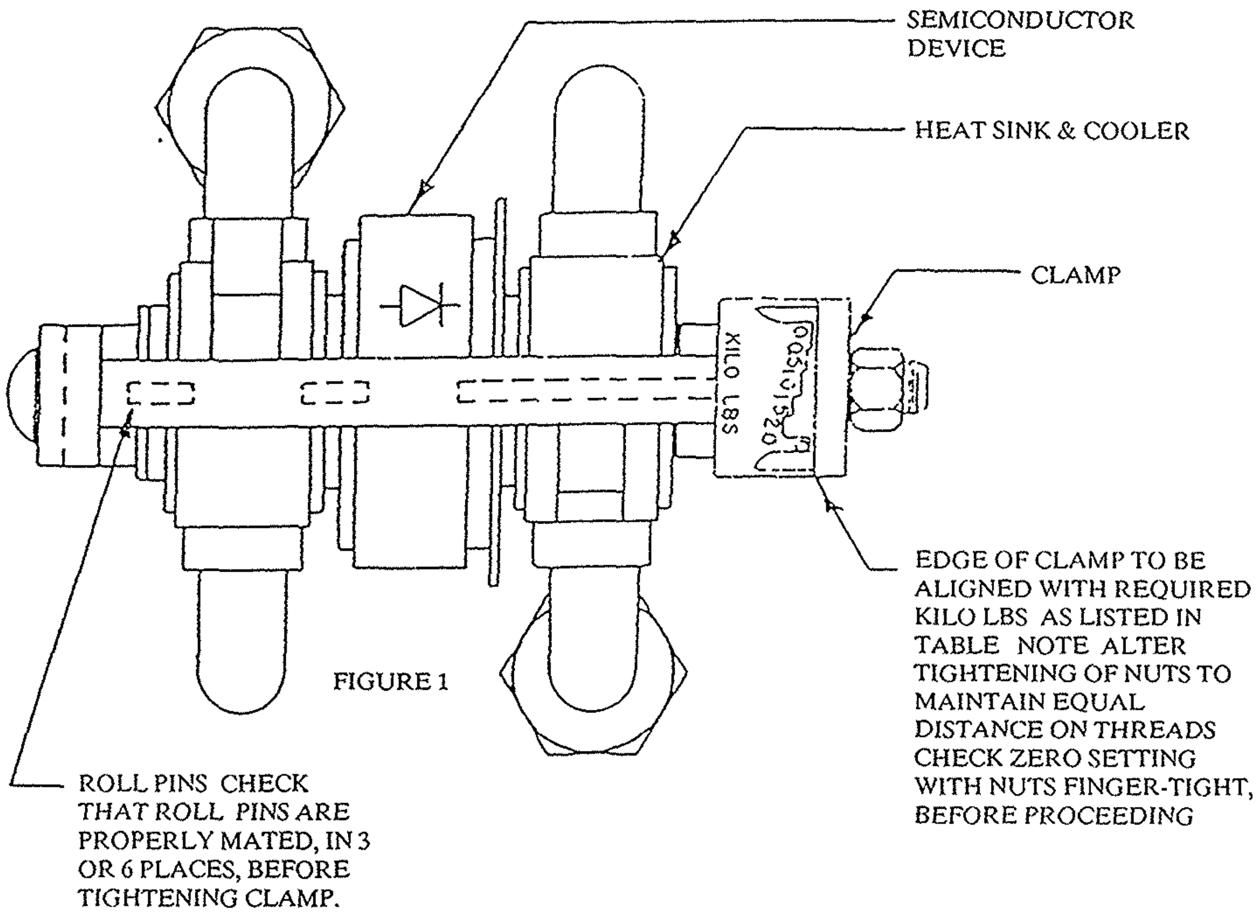


FIGURE 1

TORQUE REQUIREMENTS

SEMICONDUCTOR DEVICE	CLAMPING FORCE	
DIODE	2 KILO (2000 lbs)	
THYRISTOR SCR	2 KILO (2000 lbs)	2 DEVICES

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D. Torquing Chart (Capacitors with Studs)

CAUTION:

A. Semiconductors are not to be re-torqued or changed. See corrective maintenance for semiconductor torquing in the event of failure.

B. The terminals on capacitors are large and appear sturdy. Unfortunately, this fills the need for good electrical conductivity but is misleading mechanically. The Table of recommended maximum torque values below represents considerably less torque than one would be inclined to use on terminals of the size shown.

Type of Cap	**Stud Size**	**Max. Torque**
GE AIR COOLED	#12	25 IN/LB
GE WATER COOLED	½” (BUSHING)	180 IN/LB
GE WATER COOLED	3/8” (GROUND BAR)	180 IN/LB
WESTINGHOUSE	½” (BUSHING)	300 IN/LB
WESTINGHOUSE	3/8” (GROUND BAR)	300 IN/LB

C. See Section on Adjusting Furnace Selector Switches before Re-torquing.

*** For Ft.-Lb. – Divide In-Lbs by 12**

TORQUE (IN-LBS) *

BOLT OR SCREW SIZE	BRASS	SILICON BRONZE	STEEL
5-40	6.3	7.1	7.3
6-32	7.9	8.9	9.1
8-32	16.2	18.4	18.6
10-24	18.6	21.2	21.4
¼-20	61.5	68.8	70.9
5/16-18	107	123	124
3/8-16	192	219	222
7/16-14	317	349	354
½-13	422	480	488
9/16-12	558	632	642
5/8-11	907	1030	1044
¾-10	1249	1416	1424

E. Trouble-shooting Tests

1. General

The “trouble-shooting” section of this manual is written to facilitate maintenance using simple hand tools and a multimeter, such as a Simpson Model 260 or Triplet Model 360. Although the equipment comes from the factory with all adjustments preset and sealed, complete adjustment requires the use of an oscilloscope. As most installations using this equipment do not have an oscilloscope or people skilled in the use of one, a few simple multimeter checks and the use of operator panel meters or diagnostic board will generally locate the problem. Many times a mere visual inspection will locate a source of trouble, but an ohmmeter check should thereafter be made to verify all parts are good before re-applying power.

Advice: Do not break the seal on pots of the control board and adjust in trouble-shooting. The control can be misaligned such that further damage to the system will result and complete set up will be required. Avoid the temptation to turn pots in diagnosing a problem. As previously stated, the majority of problems can be diagnosed by use of a multimeter, panel meters and the diagnostic board.

2. Basic Power Circuit Check

These tests should be performed before applying power upon unit installation or after a failure has occurred. If fuses are found to be bad, a power semiconductor check should be made. Making these checks will be time well spent rather than gambling that everything is proper in applying power to the system.

Caution: Any of these checks are to be performed with the power supply de-energized. One should insure that no charge remains on capacitor studs or bus bars before handling.

a. Fuses – Check the power fuses with the multimeter conditioned for an ohmmeter test on the R x 1 scale. These are located at the output of the disconnect switch for incoming three phase line voltage. A good fuse will read zero ohms. A defective fuse will read more than 1000 ohms. Because of sneak paths through other components in the system, a reading other than zero ohms should be interpreted as defective.
The various control fuses will usually be good when checked after most fault conditions. If necessary, they can easily be checked by removing them from their respective sockets. If removed, insure they are returned properly. Control fuses will usually relate to obvious problems, such as no control lights, no voltmeter reading (potential transformer fuses) or lack of operator control with the power dial.

b. Power Semiconductors – The power devices (diodes and SCRs) can be checked without removing them from their heat sinks. This is advised in all cases where possible. Each should be checked in the forward and reverse direction as indicated in the test procedure for checking semiconductors. Parallel circuit paths will cause the meter to read an ohm value in one direction on good devices in some instances. A zero ohm reading (short) across a device in both directions is an improper reading. Power devices of this nature usually fail short circuit.
   CAUTION: Where devices are in parallel, it may be another part in the system that is bad, which will reflect a short on the device being measured. If a short is observed, the branch circuits should be isolated until a bad device is located.
   ANOTHER WORD OF CAUTION: When a diode is removed from the clamp, it may appear open (a high reading in both directions) which is caused by no pressure applied to the pole faces of the devices. Avoid removing devices from heat sinks until you are positive the device is damaged.
c. Capacitors – If a capacitor develops a short, it will usually read near or at zero ohms on the ohmmeter check. Again, sneak paths of parallel circuitry should be watched for when making checks of this nature. If switching leads or polarity of the multimeter indicates a different reading on the multimeter (one high – one low), the capacitor is most likely good. On a high resistance scale of the multimeter, a good capacitor will slowly charge through the batteries of the multimeter. If a short is observed on the meter and capacitors are paralleled, it will be necessary to isolate the individual section for the short. This should be verified before removing the part. A swollen capacitor in a group would be an obvious candidate for failure.

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